REAL-TIME DEEP LEARNING FRAMEWORK FOR QUANTITATIVE CANCER CELL DEFORMATION ANALYSIS IN DUAL-BEAM OPTICAL TWEEZER SYSTEMS

by

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ABSTRACT

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The mechanical behavior of cancer cells, particularly their deformation under externally applied forces, has emerged as a critical biomarker for assessing disease progression, metastatic potential, and cellular heterogeneity. Optical trapping systems, especially dual-beam optical tweezers, provide a non-invasive platform for inducing controlled deformation in single cells while capturing high-resolution image sequences. However, existing approaches for analyzing such deformation rely heavily on manual or semi-automated image processing techniques, which are time-consuming, sensitive to noise, and unsuitable for real-time applications. Furthermore, current deep learning-based methods are often constrained by limited datasets, centralized cell positioning assumptions, and lack of integration into deployable systems.

This research proposes a **real-time deep learning framework for quantitative cancer cell deformation analysis** using optical trapping imaging. The framework integrates advanced image processing, deep learning-based feature extraction, and system-level deployment into a unified pipeline. Initially, video streams acquired from the optical trapping setup are processed into high-resolution image frames, followed by preprocessing steps including noise reduction, histogram equalization, morphological operations, normalization, and resizing. A baseline convolutional neural network (CNN) is developed to predict key deformation-related parameters such as aspect ratio, area, perimeter, eccentricity, and solidity directly from cell images.

To address the limitations of conventional segmentation techniques, the proposed work incorporates a **U-Net-based semantic segmentation model** to achieve precise pixel-level delineation of cell boundaries. This enhances the accuracy and consistency of feature extraction, particularly in scenarios involving off-center cells, varying illumination conditions, and complex backgrounds. The U-Net model is trained using a combination of MATLAB generated ground-truth masks and augmented datasets to improve robustness and generalization.

Beyond single-output prediction, the framework extends to a **multi-output regression model** capable of simultaneously estimating multiple biomechanical parameters, enabling a comprehensive characterization of cell deformation. These predicted parameters are further analyzed through a biomechanical interpretation module to assess deformation trends and support classification or decision-making processes.

To ensure practical applicability, the entire system is designed for **real-time deployment and integration**, including packaging as a Python library for distribution via PyPI, and compatibility with experimental control systems. The framework supports automated data acquisition, inference, and visualization, making it suitable for high-throughput and in-situ biomedical analysis.

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1. INTRODUCTION

Modern optical tweezers systems are fundamentally opto-electromechanical instruments: laser power, electronic control, and image-based computation work together to apply precise forces and quantify cellular deformation. Dual-beam and related stretching configurations use two counter-propagating, tightly focused beams to trap and deform single cells, with deformation increasing monotonically with optical power [1]. By calibrating trap stiffness and force in the pico- to nano-Newton range, these systems provide a direct mapping from electrical drive parameters (laser current, power setpoints, AOM/galvo voltages) to the mechanical loading experienced by the cell [2].

On the sensing side, tweezers experiments are tightly integrated with high-resolution microscopy. Cell deformation is recorded as time-resolved image sequences while known forces are applied [1]. Traditional pipelines extract geometric descriptors—contours, cross-sectional area, and aspect ratio—via image/video processing and then fit mechanical models (e.g., Hertzian indentation or finite-element models) to derive stiffness and viscoelastic parameters [3]. This approach is powerful but computationally intensive and often semi-manual, limiting throughput and real-time feedback [3].

Deep learning provides a natural extension of this instrumentation. Instead of fully explicit segmentation and curve-fitting for every frame, compact convolutional neural networks (CNNs) can be trained to infer shape descriptors such as aspect ratio or deformation amplitude directly from microscopy patches, at video rate [4]. CNNs have already been shown to reliably extract coarse-grained cell shape measures from large, annotated datasets, with small hand-designed architectures outperforming heavier transfer-learning models and enabling fast training and inference on standard GPUs [4].

In parallel, encoder–decoder architectures such as U-Net enable accurate, pixel-wise segmentation of deforming cells, even under varying spatial positions and imaging conditions. These models provide precise delineation of cell boundaries, allowing robust extraction of quantitative morphological and biomechanical features [5]. By combining U-Net-based segmentation with CNN-based inference, a hybrid framework can be developed that leverages both explicit feature extraction and end-to-end learning.

This motivates a closed experimental-computational loop:

1. Apply controlled optical forces by adjusting laser power or current and trap geometry [1].
2. Acquire high-speed bright-field or phase images of the trapped, deforming cell [1].
3. Perform automated cell segmentation using a U-Net-based deep learning model to accurately delineate cell boundaries under varying deformation and spatial conditions [5].
4. Extract quantitative morphological and deformation-related features (e.g., aspect ratio, major/minor axis, and temporal deformation profiles) from the segmented cell masks [3].
5. Use a CNN-based or hybrid learning model to estimate deformation rate and maximum strain, and analyze these as functions of applied optical power and electrical drive.

1.1 Problem Statement

Most existing deformability assays rely on manual or semi-manual measurement of cell shape changes under applied forces, which is labor-intensive, operator-dependent, and limits reproducibility and throughput [6], [7], [8]. This subjectivity introduces uncertainty in the estimation of mechanical properties such as deformation rate, particularly when derived from simplified geometric assumptions. Furthermore, many conventional approaches assume linear relationships between basic morphological descriptors (e.g., area or diameter) and mechanical properties, despite the inherently non-linear and viscoelastic behavior of biological cells, leading to reduced accuracy in quantitative mechanotyping [2], [9].

Although optical and acoustic tweezer systems enable controlled, noninvasive deformation of single cells, the associated analysis pipelines are often performed offline using custom scripts and lack integration with real-time experimental control [1], [10]. In addition, existing deep learning approaches in these systems primarily focus on classification or tracking tasks, rather than directly estimating continuous deformation-related metrics or modeling deformation dynamics under varying experimental conditions [11], [12]. These limitations hinder the development of scalable, automated, and quantitatively accurate frameworks for studying cell mechanics.

Therefore, there is a need for an integrated deep learning framework that combines robust image segmentation, quantitative feature extraction, and regression-based modeling to directly estimate deformation-related parameters from optical tweezer images. Such a framework should enable accurate modeling of deformation dynamics as functions of applied optical power and electrical drive current, while supporting real-time analysis and closed-loop integration with dual-beam optical tweezer systems. Addressing this gap will enable high-throughput, operator-independent, and physically meaningful characterization of cellular mechanical properties.

1.2 Research Objectives

The primary aim of this research is to develop a real-time deep learning framework for quantitative analysis of cancer cell deformation in a dual-beam optical tweezer system. This framework integrates advanced image acquisition, deep learning-based segmentation, feature extraction, and regression modeling to characterize the mechanical response of cancer cells under controlled optical forces. The specific objectives of this study are as follows:

- To acquire and preprocess high-resolution optical tweezer images of trapped cancer cells with increased pixel sizes and variable trap positions.
- To develop and train a U-Net-based deep learning model for accurate pixel-wise segmentation of trapped cells under varying spatial positions and imaging conditions.
- Extract quantitative morphological and deformation-related features (e.g., aspect ratio, strain, and deformation index) from the segmented cell masks to serve as ground-truth labels for training a CNN-based regression model.
- To design and train a multi-output CNN-based regression model that analyzes segmented (masked or cropped) cell images to predict deformation-related metrics.

- To compute deformation rate and model its relationship with applied optical parameters, enabling quantitative analysis of deformation dynamics as functions of laser power and electrical drive current to disticguish between cancerous and non-cancerous cells.
- To implement the complete framework as a reusable Python library and publish it on PyPI for easy installation and reuse.
- To integrate the segmentation and regression models into a real-time framework coupled to the dual-beam optical tweezer setup for closed-loop monitoring and control of cell deformation.

1.3 Research Questions

To guide the development and evaluation of the proposed framework, the following research questions are formulated. These questions focus on the effectiveness of deep learning methods for segmentation and regression, as well as the ability to model and interpret cell deformation dynamics under varying experimental conditions:

- How does varying optical power and electrical drive current influence the deformation dynamics of trapped cancer cells as quantified from high-resolution optical tweezer images?
- Can a U-Net-based deep learning model accurately segment trapped cells under varying spatial positions, pixel resolutions, and imaging conditions?
- Can CNN reliably estimate quantitative deformation-related features (e.g., aspect ratio, strain, deformation index) from segmented optical tweezer images?
- How accurately can deformation rate be modeled as a function of applied optical parameters using CNN-predicted deformation metrics?
- How can the proposed segmentation and regression framework be implemented as a reusable Python library and integrated into a real-time control system for automated optical tweezer experiments?

1.4 Research Contributions

This work makes several key contributions toward the development of a data-driven framework for quantitative analysis of cancer cell deformation in optical tweezer systems.

- Development and training of a U-Net-based deep learning model for robust and accurate pixel-wise segmentation of trapped cancer cells under varying spatial positions, imaging resolutions, and deformation conditions.
- Establishment of a quantitative feature extraction pipeline that derives morphological and deformation-related metrics (e.g., aspect ratio, strain, and deformation index) from U-Net-generated segmentation masks for use as ground-truth labels.
- Design, training, and validation of a multi-output CNN-based regression model capable of predicting deformation-related metrics directly from segmented optical tweezer images.
- Formulation of a data-driven approach for modeling deformation dynamics by relating predicted deformation metrics to applied optical parameters, including laser power and electrical drive current.
- Implementation of the complete segmentation and regression pipeline as a reusable and well-documented Python library to facilitate reproducibility and adoption by the research community.
- Integration of the proposed framework into a real-time computational pipeline for continuous monitoring and analysis of cell deformation in dual-beam optical tweezer systems.
- Provision of a generalizable methodology for combining optical manipulation, deep learning-based segmentation, and regression modeling for quantitative biomechanical characterization of biological cells.

2. BACKGROUND

2.1 Cancer Cell Mechanics and Deformation

Cancer cell mechanics play a fundamental role in understanding disease progression and metastatic potential. Mechanical properties such as stiffness, elasticity, viscosity, and viscoelasticity govern how cells respond to external forces and interact with their surrounding microenvironment. These properties influence essential biological processes including migration, circulation, proliferation, and tissue invasion, and are strongly linked to pathological conditions such as cancer and fibrosis [13]. Notably, cancer cells often exhibit altered mechanical behavior compared to normal cells, typically becoming more deformable as malignancy increases.

Cell deformation refers to the change in shape of a cell when subjected to external forces. This deformation provides quantitative insight into the structural and biomechanical properties of the cell, including cytoskeletal organization, membrane integrity, and intracellular composition. From a mechanical perspective, these properties can be described using classical constitutive relations.

Elasticity, often quantified by Young's modulus, describes the resistance of a cell to deformation under applied stress and is defined as

$$E = \frac{\sigma}{\varepsilon} \tag{1}$$

where E is the Young's modulus, $\sigma = \frac{F}{A}$ represents the applied stress, and $\varepsilon = \frac{\Delta L}{L_0}$ is the resulting strain. In biological systems, a lower Young's modulus corresponds to increased deformability, a characteristic frequently associated with malignant cells [14].

In addition to elastic behavior, cells exhibit viscous properties, which describe their time-dependent response to applied forces. This behavior is characterized by gradual deformation (creep) and stress relaxation, indicating that cells do not respond instantaneously but evolve over time under mechanical loading [13], [15].

More realistically, cells behave as viscoelastic materials, exhibiting both solid-like and fluid-like characteristics. This combined behavior is commonly modeled using constitutive equations such as the Kelvin–Voigt model:

$$\sigma(t) = E\,\varepsilon(t) + \eta\,\frac{d\varepsilon(t)}{dt} \tag{2}$$

where η represents the viscosity coefficient. Furthermore, experimental studies have shown that cellular deformation often follows a power-law relationship over time:

$$J(t) \propto t^{\alpha} \tag{3}$$

where $J(t)$ is the creep compliance and α is the power-law exponent, reflecting the complex and heterogeneous internal structure of cells [15], [16].

In practical applications, deformation is often quantified using geometric descriptors derived from imaging data. A commonly used metric is the deformation index:

$$D = \frac{a-b}{a+b} \tag{4}$$

where a and b represent the major and minor axes of the deformed cell, respectively. Such descriptors provide a direct link between observed morphology and underlying mechanical properties, forming the basis for automated analysis using computational and deep learning approaches.

2.1.1 Relevance to Disease Progression. The mechanical properties of cancer cells are increasingly recognized as functionally relevant biomarkers of disease progression rather than merely passive physical descriptors. During tumor development, cancer cells undergo cytoskeletal remodeling, altered cell–cell and cell–extracellular matrix (ECM) adhesion, and changes in membrane and nuclear mechanics. These transformations modify cellular stiffness, deformability, and viscoelastic response, thereby influencing proliferation, invasion, intravasation, circulation, and colonization of distant tissues [17], [18], [19].

A widely observed hallmark of malignancy is the reduction in cellular stiffness. Mechanically, this corresponds to a decrease in Young's modulus E, leading to increased deformability. Such deformability enables cancer cells to traverse confined environments such as dense ECM networks, basement membranes, and microvasculature, thereby facilitating metastatic dissemination [18], [19], [20]. For instance, studies on ovarian cancer have demonstrated that highly metastatic cells exhibit significantly lower stiffness compared to less aggressive cells, establishing mechanical phenotype as a predictive biomarker of metastatic potential [20]. Similarly, optical deformability measurements have shown that malignant cells deform more readily than normal cells, reinforcing the role of biomechanics as an intrinsic indicator of disease state [21].

Beyond stiffness, viscoelastic properties play a critical role in enabling cancer cells to adapt to dynamic mechanical stresses within the tumor microenvironment. Tumor tissues are characterized by elevated interstitial pressure, ECM stiffening, and fluctuating mechanical forces. Under such conditions, the time-dependent deformation behavior described by viscoelastic models allows cells to withstand, dissipate, and recover from mechanical stress, thereby enhancing survival and persistence during metastasis [17], [18], [22].

From a clinical perspective, these mechanical signatures provide a powerful, label-free means of distinguishing between normal, benign, and malignant phenotypes. Techniques such as atomic force microscopy, optical stretching, microfluidic deformability cytometry, and optical trapping-based deformation analysis enable precise quantification of these properties [19], [20], [21]. Importantly, the integration of quantitative deformation metrics with imaging and computational models opens new avenues for automated diagnosis, staging, and real-time monitoring of cancer progression.

Overall, cancer cell mechanics offer critical insight into the physical mechanisms underlying metastasis and tumor evolution. The strong correlation between mechanical phenotype and disease progression underscores the importance of quantitative deformation analysis as both a diagnostic tool and a foundation for advanced computational frameworks in biomedical research [17], [18], [22].

2.1.2 Measurement Techniques. Cell mechanical properties can be measured through several sophisticated techniques, each offering unique advantages for specific applications. Atomic Force Microscopy (AFM) is a widely used technique for quantifying the mechanical properties of biological cells at micro- and nanoscale resolution. As shown in Figure (1), a flexible cantilever with a sharp probe tip indents the surface of a cell, and the resulting cantilever deflection is measured using an optical lever system consisting of a laser beam reflected onto a position-sensitive photodiode [23]. The measured deflection is converted into force, enabling precise characterization of the interaction between the probe and the sample.

During indentation, a force–indentation curve is obtained and analyzed using contact mechanics models, such as the Hertz model, to estimate the Young's modulus of the cell:

$$F = \frac{4}{3} E^* \sqrt{R}\, \delta^{3/2} \tag{5}$$

where F is the applied force, E^* is the effective elastic modulus, R is the tip radius, and δ is the indentation depth [24].

AFM has been extensively applied in cancer research, where studies have shown that malignant cells are typically softer than normal cells, highlighting its potential as a diagnostic biomarker [25]. However, AFM is inherently low-throughput, requires physical contact with the sample, and is sensitive to experimental conditions. Furthermore, it lacks real-time processing capabilities, limiting its applicability in high-throughput and automated diagnostic systems.

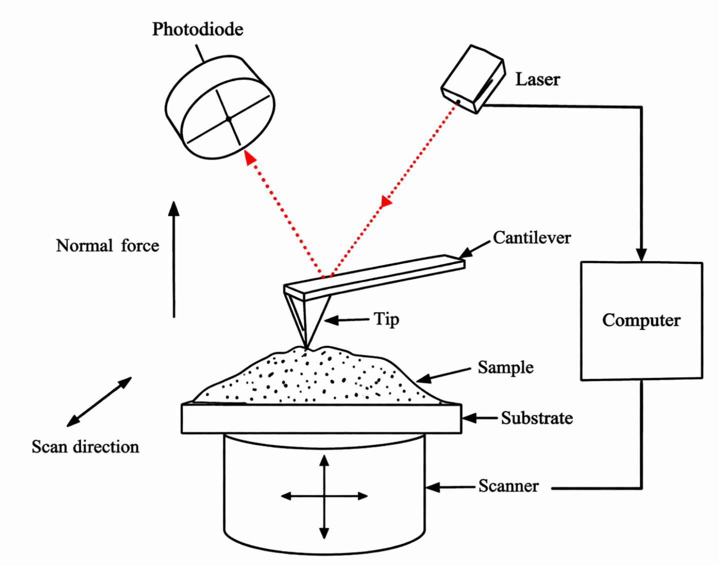

Figure 1: Schematic diagram of atomic force microscopy (AFM) showing cantilever-based indentation and optical detection.

In addition, acoustic beam tweezers use focused ultrasound waves to trap, manipulate, and mechanically interrogate biological cells without physical contact [6]. In these systems, an acoustic transducer generates a pressure field within the sample medium, and the resulting acoustic radiation force acts on suspended particles or cells. By properly controlling the acoustic frequency, amplitude, and beam profile, cells can be translated, trapped, or deformed in a controlled manner. Unlike optical tweezers, acoustic beam tweezers can operate effectively in optically opaque environments and generally produce lower photothermal effects, making them attractive for biological and biomedical applications [6].

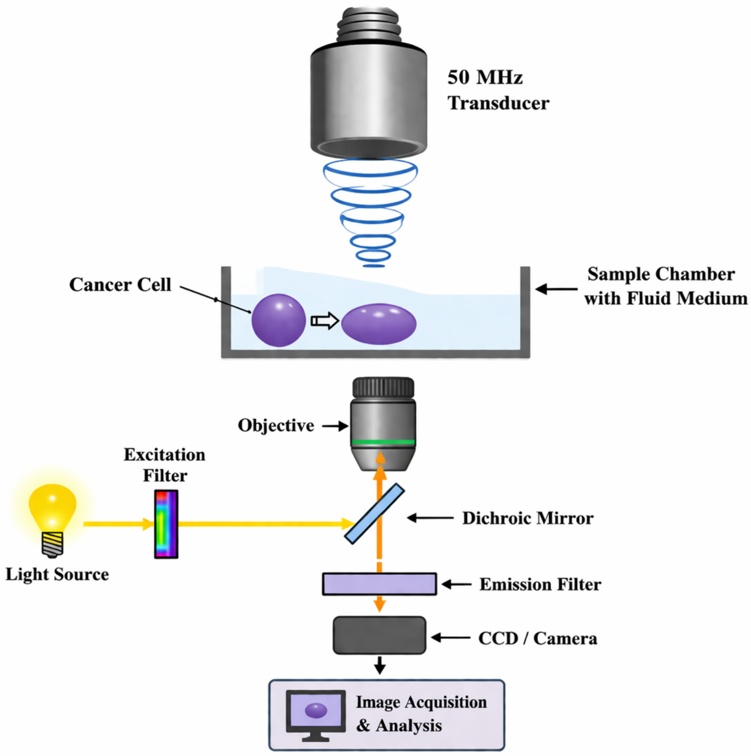

Figure 2: Schematic of a single-beam acoustic tweezer (SBAT) system showing a focused ultrasound transducer for cell manipulation and an integrated optical imaging pathway for deformation observation and analysis.

For cell mechanics studies, acoustic beam tweezers provide a non-invasive means of applying mechanical stress while imaging systems record the resulting deformation. The captured images can then be analyzed to extract morphological descriptors such as aspect ratio and deformation index, which are used to characterize the cell's biomechanical response. Because of their contactless operation, compatibility with microfluidic platforms, and suitability for real-time imaging, acoustic beam tweezers serve as an important measurement technique for quantitative cell deformation analysis.

2.2 Dual-Beam Optical Tweezer Systems

Dual-beam optical tweezers are widely used for the non-contact manipulation and mechanical characterization of biological cells. The technique relies on the interaction between focused laser beams and dielectric particles, where optical forces arise from momentum transfer due to light refraction and scattering [26]. In a dual-beam configuration, two counter-propagating laser beams create a stable trapping region that confines the cell at the center while allowing controlled mechanical loading [1]. By increasing the laser intensity, the opposing forces stretch the cell along the beam axis, enabling precise and symmetric deformation.

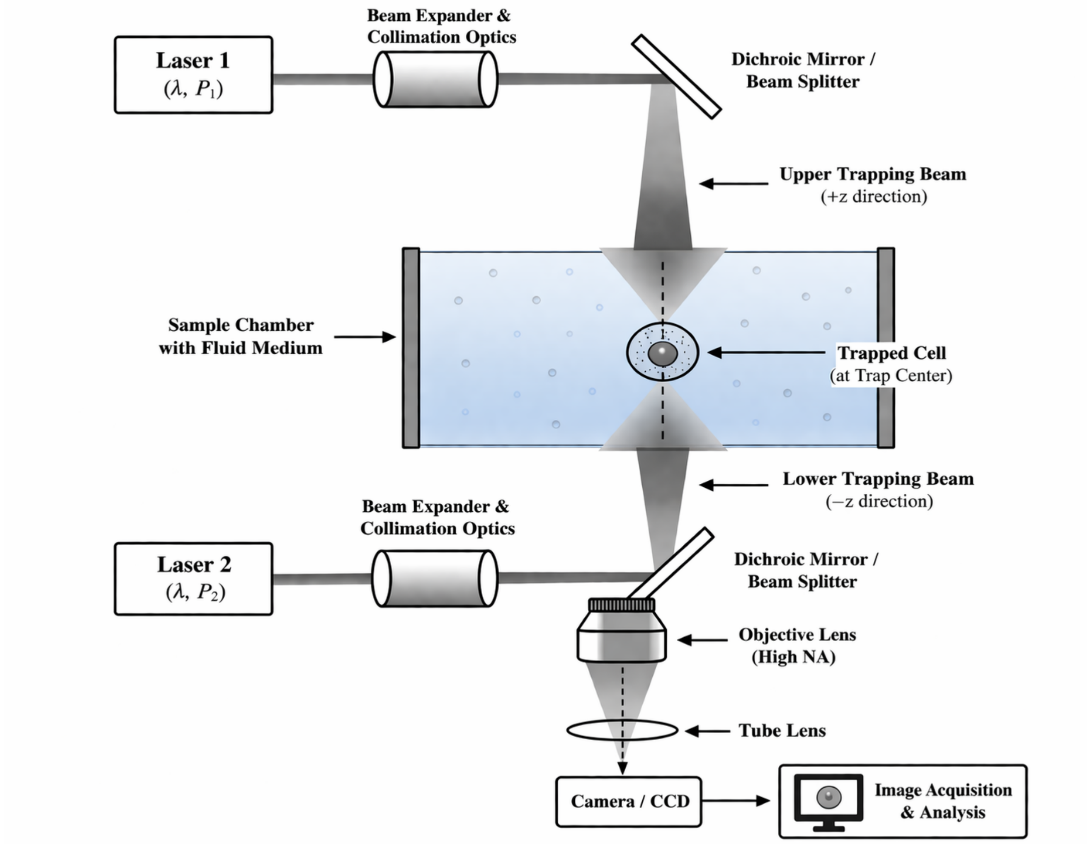

Figure 3: Schematic of a dual-beam optical tweezer system showing counter-propagating laser beams trapping a cell at the center of a fluid medium and enabling controlled deformation within the sample chamber.

The deformation process is monitored using optical microscopy, allowing real-time observation of cell shape changes. Quantitative descriptors such as aspect ratio and deformation index are extracted from the captured images to characterize cellular mechanical properties. Compared to contact-based techniques such as atomic force microscopy, dual-beam optical tweezers provide non-invasive manipulation, reduced risk of damage, and compatibility with controlled microenvironments. These advantages, combined with recent advances in optical manipulation and biomedical applications, make the technique well suited for integration with computational methods, including deep learning-based frameworks for automated and real-time deformation analysis.

2.2.1 Imaging in Optical Trapping Systems. Imaging plays a critical role in optical trapping systems by enabling real-time observation and quantitative analysis of cell deformation under applied optical forces. In these systems, cells are manipulated using focused laser beams, while high-resolution microscopy techniques such as bright-field or phase-contrast imaging capture their dynamic morphological changes. As the trapped cell deforms, its shape evolution is recorded as a sequence of images, providing a direct representation of its mechanical response.

Quantitative analysis is performed by extracting morphological features such as area, perimeter, aspect ratio, and deformation index, defined in Equation (4), where a and b represent the major and minor axes of the deformed cell, respectively. While this image-based approach enables non-invasive and real-time analysis and is well suited for integration with machine learning frameworks, traditional methods rely on manual segmentation and handcrafted features, which are sensitive to noise and imaging variability. These limitations motivate the use of deep learning-based approaches for robust and automated deformation analysis.

2.3 Deep Learning for Biomedical Image Analysis

Deep learning has emerged as a powerful approach for biomedical image analysis, enabling automated extraction of complex and high-level features directly from raw image data [12], [27]. Unlike traditional image processing techniques that rely on handcrafted features and manual segmentation, deep learning models, particularly CNNs can learn hierarchical representations through multiple layers of nonlinear transformations. This capability allows them to capture intricate spatial patterns and variations in biological structures, making them highly effective for tasks such as image classification, segmentation, and object detection.

In the context of cellular imaging, deep learning has significantly improved the accuracy and robustness of analyzing morphological changes under varying experimental conditions [28]. CNN-based architectures, such as U-Net, have demonstrated strong performance in biomedical image segmentation, enabling precise delineation of cell boundaries even in noisy or low-contrast images. Furthermore, deep learning models can generalize across variations in illumination, cell positioning, and imaging modalities, addressing key limitations associated with traditional feature extraction methods.

The integration of deep learning with optical trapping systems provides a framework for automated and real-time analysis of cell deformation. Instead of relying solely on predefined geometric descriptors, deep learning models can learn discriminative features that capture both linear and nonlinear deformation characteristics. This enables more robust classification and quantitative assessment of cellular mechanical properties, supporting the development of intelligent and scalable biomedical analysis systems.

2.3.1 Convolutional Neural Networks. CNNs are designed to process data that come in the form of multiple arrays, for example a color image composed of three 2D arrays containing pixel intensities in the three color channels [29]. CNNs are a class of deep learning models specifically designed for processing grid-like data such as images. They are particularly effective in biomedical image analysis due to their ability to automatically learn spatial hierarchies of features through convolutional operations [30]. Unlike traditional machine learning approaches that rely on handcrafted features, CNNs learn feature representations directly from raw pixel intensities, enabling improved robustness and generalization.

As shown in Figure (4), a typical CNN architecture consists of multiple layers, including convolutional layers, activation functions, pooling layers, and fully connected layers. Convolutional layers apply learnable filters to the input image to extract local features such as edges, textures, and shapes [31]. These features are then passed through nonlinear activation functions, such as the Rectified Linear Unit (ReLU), which introduce nonlinearity and improve model expressiveness. Pooling layers are used to reduce the spatial dimensions of feature maps, thereby decreasing computational complexity and providing translational invariance.

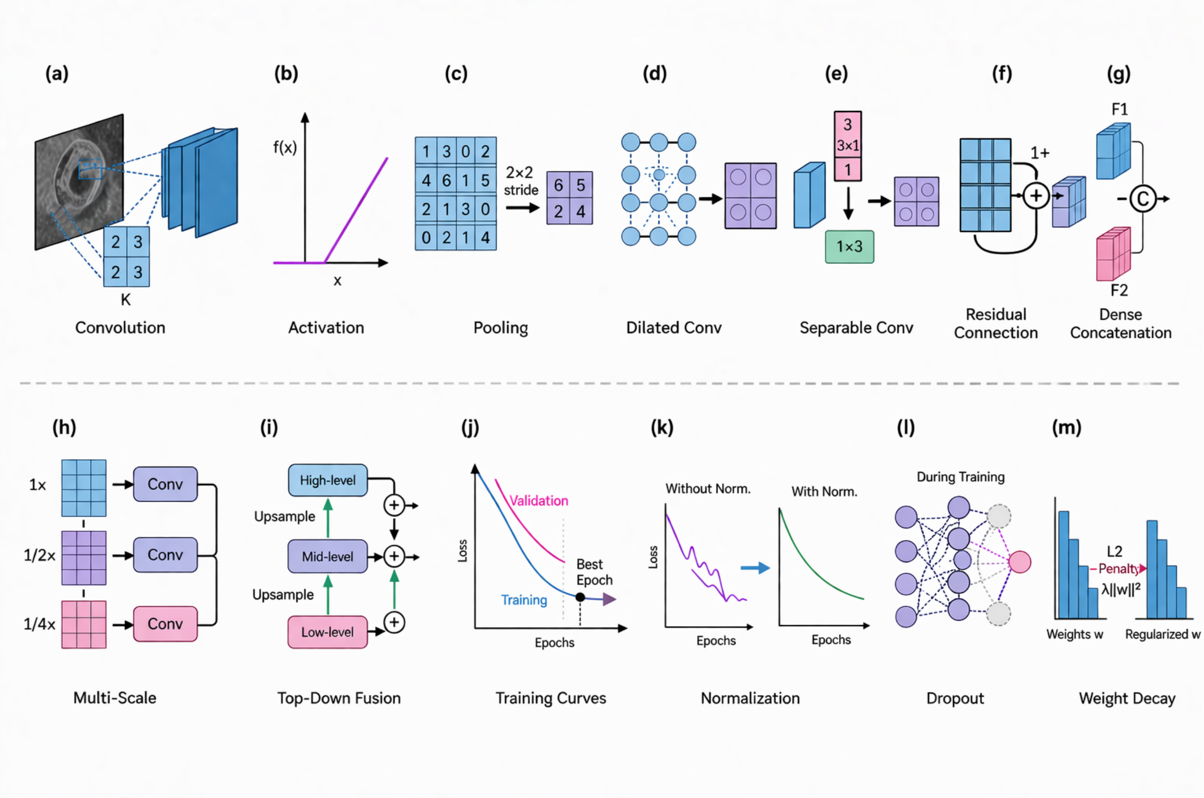

Figure 4: Fundamental components of convolutional neural networks. (a) Convolutional operations extract local spatial features using learnable filters. (b) Activation functions, such as the rectified linear unit (ReLU), introduce nonlinearity to enable complex function approximation. (c) Pooling operations reduce spatial resolution while preserving dominant features, producing compact feature maps. Repeated application of these operations enables hierarchical feature learning. (d) Dilated convolutions expand the receptive field without increasing the number of parameters, allowing broader contextual information to be captured. (e) Separable convolutions reduce computational cost by decomposing standard convolutions into channel-wise operations. (f) Residual connections facilitate the training of deep networks by learning residual mappings. (g) Dense connections promote feature reuse by linking each layer to all preceding layers. (h) Multi-scale representations enable the model to capture both fine and coarse features. (i) Feature fusion mechanisms combine information from different resolutions to enhance representation capability. (j) Training curves illustrate model behavior, highlighting underfitting and overfitting as functions of model capacity. (k) Normalization techniques improve training stability and convergence. (l) Dropout randomly deactivates neurons during training to reduce overfitting. (m) Weight regularization penalizes large parameter values, encouraging better generalization.

As the network depth increases, CNNs learn increasingly abstract representations of the input data. Early layers capture low-level features such as edges and contours, while deeper layers encode high-level semantic information relevant to the task. In the context of cell deformation analysis, these hierarchical features enable the model to capture subtle morphological variations that may not be easily described using traditional geometric descriptors.

The final layers of a CNN typically consist of fully connected layers or global pooling operations, which aggregate the learned features and map them to the desired output, such as classification or regression. For deformation analysis, the network can be designed to predict deformation metrics or classify cells based on their mechanical properties. This end-to-end learning framework allows CNNs to automatically extract and optimize relevant features, making them highly suitable for real-time and automated biomedical image analysis [31].

2.3.2 U-Net for Image Segmentation. U-Net is a CNN specifically designed for biomedical image segmentation. It was introduced to address the challenge of accurate pixel-level prediction in scenarios with limited annotated data[5]. The architecture follows an encoder–decoder structure, where the encoder progressively captures contextual information through convolution and pooling operations, while the decoder reconstructs spatial details through upsampling.

A key feature of U-Net is the use of skip connections between corresponding layers in the encoder and decoder paths. These connections allow high-resolution features from the encoder to be combined with upsampled features in the decoder, improving localization accuracy and preserving fine structural details. This design enables precise segmentation of objects with complex boundaries, such as biological cells [32].

The network architecture is illustrated in Figure (6). It consists of a contracting path (left side) and an expansive path (right side). The contracting path follows the typical architecture of a convolutional network. It consists of the repeated application of two 3x3 convolutions (unpadded convolutions), each followed by a rectified linear unit (ReLU) and a 2x2 max pooling operation with stride 2 for downsampling. At each downsampling step we double the number of feature channels. Every step in the expansive path consists of an upsampling of the feature map followed by a 2x2 convolution ("up-convolution") that halves the number of feature channels, a concatenation with the correspondingly cropped feature map from the contracting path, and two 3x3 convolutions, each followed by a ReLU. The cropping is necessary due to the loss of border pixels in every convolution [5], [33]. At the final layer a 1x1 convolution is used to map each 64-component feature vector to the desired number of classes.

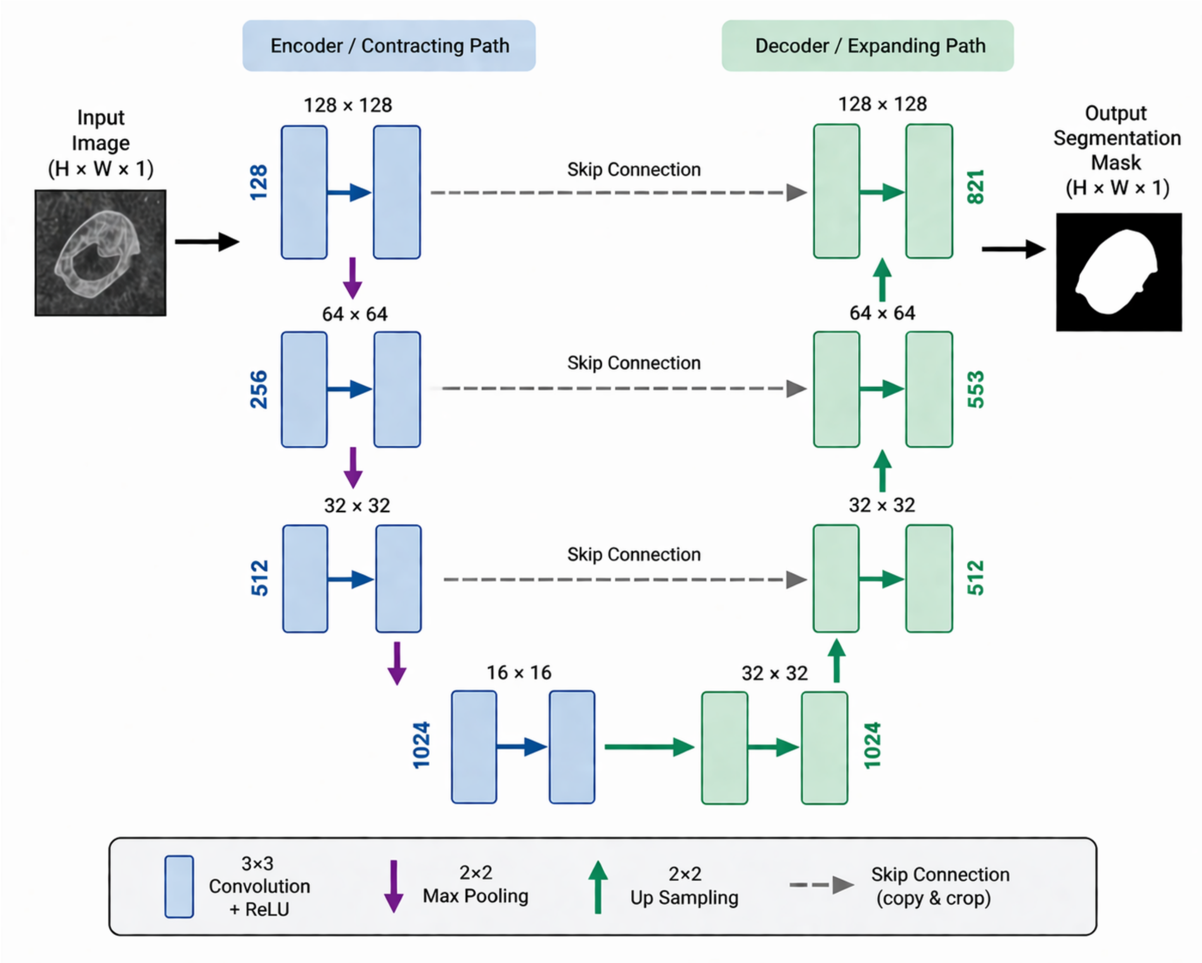

Figure 5: U-Net network architecture, showing the encoder (contracting path), bottleneck, and decoder (expanding path) with skip connections that transfer high-resolution features through copy and crop operations to improve localization accuracy.

Overall, the ability of U-Net to perform end-to-end learning for pixel-wise classification makes it a powerful tool for biomedical image analysis and a foundational component in deep learning-based frameworks for quantitative deformation analysis [33].

2.4 Morphological Feature Extraction

Morphological feature extraction is a fundamental step in the quantitative analysis of cell deformation, where geometric properties are derived from imaging data to characterize the mechanical response of cells. In optical trapping systems, captured images of deformed cells are processed to extract meaningful descriptors that reflect changes in cell shape and structure.

The process typically begins with image preprocessing and segmentation, where the cell boundary is identified using techniques such as thresholding, edge detection, or contour extraction. Once the cell is segmented, various morphological features can be computed.

Commonly used features include the cell area and perimeter, which describe the size and boundary characteristics of the cell. Shape descriptors such as aspect ratio and circularity provide additional insight into deformation. The aspect ratio is defined as the ratio of the major axis to the minor axis of the cell:

$$AR = \frac{a}{b} \tag{6}$$

where a and b represent the lengths of the major and minor axes, respectively.

Another important metric is the deformation index, which quantifies the extent of shape distortion as shown in equation (4). These features provide a direct link between observed morphology and underlying mechanical properties, enabling quantitative comparison between normal and diseased cells.

Despite their effectiveness, traditional morphological feature extraction methods rely heavily on accurate segmentation and handcrafted feature design [34]. These approaches are often sensitive to noise, variations in illumination, and changes in cell position or orientation. Furthermore, they may fail to capture complex, non-linear patterns present in deformation behavior. These limitations motivate the use of deep learning-based approaches, which can automatically learn robust and discriminative features directly from raw image data.

2.5 Related Works

Recent studies have increasingly explored the integration of physical cell manipulation platforms with deep learning to enhance cancer-cell analysis and diagnostics. In one of the early efforts in this direction, Lim et al. [7] combined high-frequency Single-Beam Acoustic Tweezers (SBAT) with CNNs to classify breast cancer cells based on deformation patterns. Their framework exploited the relationship between cell elasticity and metastatic potential, achieving precision and recall values above 0.95. Despite this promising performance, the study relied on a relatively small dataset collected under tightly controlled imaging conditions with minimal spatial variability in cell positioning, limiting the robustness and scalability of the approach for real-time automated analysis.

Building upon this idea, Lee et al. [6] moved beyond categorical classification toward quantitative biomechanical characterization. They proposed a hybrid CNN–MLP framework in which a CNN estimated area-change ratios of cells subjected to varying acoustic pressures, while a multilayer perceptron mapped these deformation features to a scalar elastic modulus. By eliminating manual boundary tracing and enabling automated deformability estimation, the work demonstrated the potential of deep learning to infer biomechanical properties from deformation data. However, the framework remains specific to acoustic manipulation platforms and does not address broader challenges such as flexible imaging conditions, positional variability, or integration into fully automated analysis pipelines.

While early work focused largely on acoustic tweezers, optical trapping technologies have emerged as a highly versatile alternative for manipulating microscopic particles and biological cells. Yang et al. [35] provide a comprehensive review of optical sorting systems, demonstrating how optical tweezers can be integrated with microfluidics, Raman spectroscopy, fluorescence imaging, immunoassays, and AI-based image analysis to achieve highly precise active and passive particle sorting. The study highlights significant improvements in sorting resolution, with sub-100-nm particles separable at sub-10-nm precision and theoretical limits approaching nanometer-scale resolution.

Similarly, Yadav et al. [3] review advances in optical tweezers used for biomedical applications, covering trapping principles, calibration methods, and emerging architectures such as multiple traps and acousto-optic steering. Their survey documents applications including indentation, stretching, and membrane tethering assays that quantify cellular mechanical properties, enabling differentiation between normal, cancerous, and leukemic cells. Despite these advances, optical tweezers still face limitations related to photothermal damage, reduced trapping efficiency for extremely small particles, and restricted force ranges, highlighting the need for improved system design and intelligent data analysis techniques.

To address these challenges, Ciarlo et al. [12] explore the role of deep learning in improving optical tweezer systems. Their perspective review demonstrates how neural networks can assist in tasks such as particle tracking, calibration, force estimation, and real-time control of optical traps. Reported results show that deep learning–based particle tracking can outperform classical algorithms in noisy environments, while neural-network force estimation can be faster and sometimes more accurate than traditional analytical models. However, the authors emphasize that the success of these approaches depends heavily on the availability of large and well-curated datasets, and interpretability remains a challenge.

In parallel with developments in cell manipulation systems, deep learning has significantly improved the analysis of biomedical images. CNN-based models have been widely applied for tasks such as disease detection, tissue classification, and anatomical segmentation. For example, Mary and Shanthasheela [36] proposed a hybrid framework combining morphological image processing and CNN classification for lung cancer detection from CT images. Their approach involved preprocessing and morphological segmentation followed by CNN-based classification, achieving an overall accuracy of approximately 90.32%. Although the method demonstrated high sensitivity and low misclassification rates, it relied on traditional segmentation techniques rather than modern deep learning architectures and focused solely on classification rather than quantitative analysis.

More recent studies have emphasized improving segmentation accuracy using advanced CNN architectures. Vishalakshi and Pramod Kumar [37] introduced an optimized CNN–U-Net model incorporating residual blocks and attention gates for MRI-based kidney segmentation. Their model achieved a Dice similarity coefficient of 92.8% and reduced boundary errors, demonstrating the effectiveness of architectural improvements for precise medical image segmentation. However, these models are primarily designed for segmentation tasks and do not extend to downstream prediction of quantitative metrics related to cellular deformation or mechanical properties.

To further improve diagnostic performance, several studies have explored ensemble learning approaches that combine predictions from multiple deep learning models. For instance, Eleyan et al. [38] proposed an ensemble framework integrating pretrained CNN architectures such as ResNet, DenseNet, EfficientNet, and MobileNet for brain tumor detection from MRI images. Using transfer learning and majority voting, their model achieved an accuracy of 98.5%, outperforming individual networks. The improvement was attributed to the complementary strengths of different architectures and the ability of ensemble methods to reduce model bias and variance.

Similarly, Nemade et al. [39] demonstrated that transfer learning combined with ensemble classifiers can significantly improve breast cancer classification performance. Their approach utilized pretrained CNN models including VGG16, VGG19, and InceptionV3 to extract deep features, which were subsequently combined using ensemble classifiers. While these approaches achieve high classification accuracy, they remain limited to categorical predictions and do not provide quantitative insight into morphological or biomechanical properties.

Hybrid frameworks that combine segmentation and classification have also been investigated. For example, Golkarieh et al. [40] proposed a pipeline integrating U-Net segmentation with CNN feature extraction and classical machine learning classifiers such as Support Vector Machines (SVM) and Random Forests. Their results demonstrated that deep features can effectively replace handcrafted descriptors in classification tasks. However, the framework remains limited to binary classification and does not address regression-based modeling of cellular deformation characteristics.

Recent research has also investigated architectures capable of integrating multiple data modalities. Uma Kandan et al. [41] proposed a multi-input CNN framework in which parallel network branches process different types of input data before fusing the extracted features for final prediction. This architecture allows the model to capture both spatial information from images and contextual information from structured data. Experimental results show that multimodal feature fusion can significantly improve predictive performance compared with single-input CNN models. Nevertheless, these approaches are typically applied to classification tasks and do not explicitly address the prediction of biomechanical properties or multi-output regression.

While most CNN-based models focus on classification tasks, some studies have explored regression-based frameworks for predicting continuous variables. For example, Chowdhury and Das [42] proposed a CNN-based regression model for predicting the maternal mortality ratio using socio-demographic data. Their model outperformed traditional regression methods such as linear regression, ridge regression, Lasso regression, decision trees, and random forests, achieving approximately 12% higher prediction performance. These results demonstrate the capability of CNN architectures to model complex nonlinear relationships in regression tasks.

In the biomedical imaging domain, Elkassa et al. [43] proposed a hybrid CNN framework that integrates handcrafted features such as Histogram of Oriented Gradients (HOG) and Gray-Level Co-occurrence Matrix (GLCM) with deep features extracted from OCT images to predict visual acuity. The model formulates the problem as a regression task, allowing the prediction of continuous clinical outcomes rather than categorical labels. Although the integration of deep and handcrafted features improved predictive performance, the framework is limited to the estimation of a single clinical parameter and introduces additional preprocessing complexity.

The integration of optical imaging with deep learning has also been explored for real-time clinical decision support. For instance, Sivamurugan and Sureshkumar [44] proposed a framework combining optical coherence tomography (OCT) imaging with a U-Net-based CNN to segment tumor regions during biopsy procedures. Using micron-scale OCT images, the model achieved an AUC of approximately 0.877 and demonstrated nearly 90% agreement with expert interpretations. These results highlight the potential of AI-assisted optical imaging systems for improving diagnostic accuracy and supporting clinical decision-making.

To provide a broader perspective, review studies have synthesized recent progress in applying deep learning to biomedical imaging. For example, Pérez-Núñez et al. [45]

summarize the widespread adoption of CNNs, U-Net architectures, and hybrid models for tasks such as segmentation, classification, and disease detection. Their review highlights substantial improvements in diagnostic accuracy and automation enabled by deep learning. However, the authors also note that most existing approaches focus primarily on classification and segmentation tasks, with limited attention to quantitative prediction and continuous parameter estimation. Furthermore, current models often lack the ability to integrate multiple data modalities and rarely provide physically interpretable outputs, indicating a need for more advanced frameworks capable of quantitative and multimodal biomedical image analysis.

2.6 Identified Research Gap

Despite significant advancements in deep learning for biomedical image analysis, several critical limitations remain. Existing approaches are predominantly focused on classification and segmentation tasks, where models are designed to identify or localize regions of interest rather than provide quantitative insights into underlying physical or biomechanical properties. While these methods have achieved high accuracy in disease detection, they offer limited interpretability in applications where continuous characterization of cellular behavior is required.

Furthermore, studies that attempt quantitative analysis are often constrained to single-parameter estimation or rely on hybrid pipelines that incorporate handcrafted features alongside deep learning models. Such approaches introduce additional preprocessing complexity and may not generalize well across varying imaging conditions. In addition, most existing frameworks are developed for specific imaging modalities, such as acoustic tweezers or optical coherence tomography, and do not address the challenges associated with spatial variability, dynamic cell positioning, and real-time system integration.

Another key limitation is the lack of unified architectures capable of integrating heterogeneous data sources. While recent work has explored multi-input and multimodal CNN frameworks, these approaches are primarily applied to classification problems and do not extend to multi-output regression tasks involving physically meaningful parameters. As a result, current models fail to fully exploit the complementary strengths of image-based features and structured morphological descriptors for comprehensive analysis.

Moreover, in the context of optical trapping systems, there is a notable absence of deep learning frameworks specifically designed for deformation-based analysis under realistic experimental conditions. Existing studies either focus on controlled environments with limited variability or do not incorporate the complexities associated with real-time imaging and automated decision-making.

To address these limitations, this work proposes a real-time deep learning framework for quantitative cancer cell deformation analysis using optical trapping imaging. The proposed approach integrates advanced image segmentation, feature extraction, and a multi-input regression model to enable the prediction of multiple deformation-related metrics from biomedical images. By combining spatial information from images with structured morphological features, the framework aims to provide a comprehensive and physically interpretable analysis of cellular behavior, while supporting robust performance under varying experimental conditions.

3. PROPOSED METHODOLOGY

3.1 Overview

This research proposes a real-time deep learning framework for quantitative analysis of cancer cell deformation using optical trapping imaging. The methodology is designed as a progressive and modular pipeline that integrates image acquisition, preprocessing, deep learning–based analysis, and decision-making for deformation estimation.

The overall system begins with the acquisition of cell images from an optical trapping setup, where individual cancer cells are subjected to controlled optical forces. These images are then processed through a series of preprocessing techniques, including noise reduction, contrast enhancement, and morphological operations, to improve image quality and consistency for downstream analysis.

A baseline CNN model has been developed and implemented using GPU acceleration to establish the feasibility of deep learning for deformation analysis. This model predicts the morphological features of the input images and serves as a reference point for estimating the deformation rate. However, initial observations indicate that the CNN model is limited in its ability to localize cell boundaries and accurately capture deformation characteristics, particularly in images where cells are not centrally positioned.

To address these limitations, the proposed framework incorporates a U-Net–based segmentation model for precise extraction of cell regions from the background. The segmentation output enables the computation of quantitative morphological features, such as area, perimeter, aspect ratio, and eccentricity, which provide a physically meaningful representation of cell deformation.

These extracted features are subsequently used for deformation estimation through a learning-based decision model. The integration of segmentation and feature-based analysis is expected to improve robustness, interpretability, and accuracy compared to direct CNN-based classification.

Furthermore, the framework is designed with real-time deployment in mind, allowing integration with optical trapping systems for automated and efficient analysis. The modular architecture enables future scalability and optimization, including improvements in model training, data augmentation, and inference performance.

Overall, this methodology bridges the gap between deep learning–based image analysis and quantitative biomechanical characterization of cancer cell deformation, providing a foundation for robust and deployable diagnostic tools.

3.2 System Architecture Design

The proposed system is designed as a modular deep learning framework for quantitative cancer cell deformation analysis using optical trapping images. The architecture is organized to support the complete analysis pipeline, beginning from image acquisition and ending with deformation estimation and decision output. This modular design allows each stage of the system to be independently developed, evaluated, and improved while maintaining compatibility with the overall workflow.

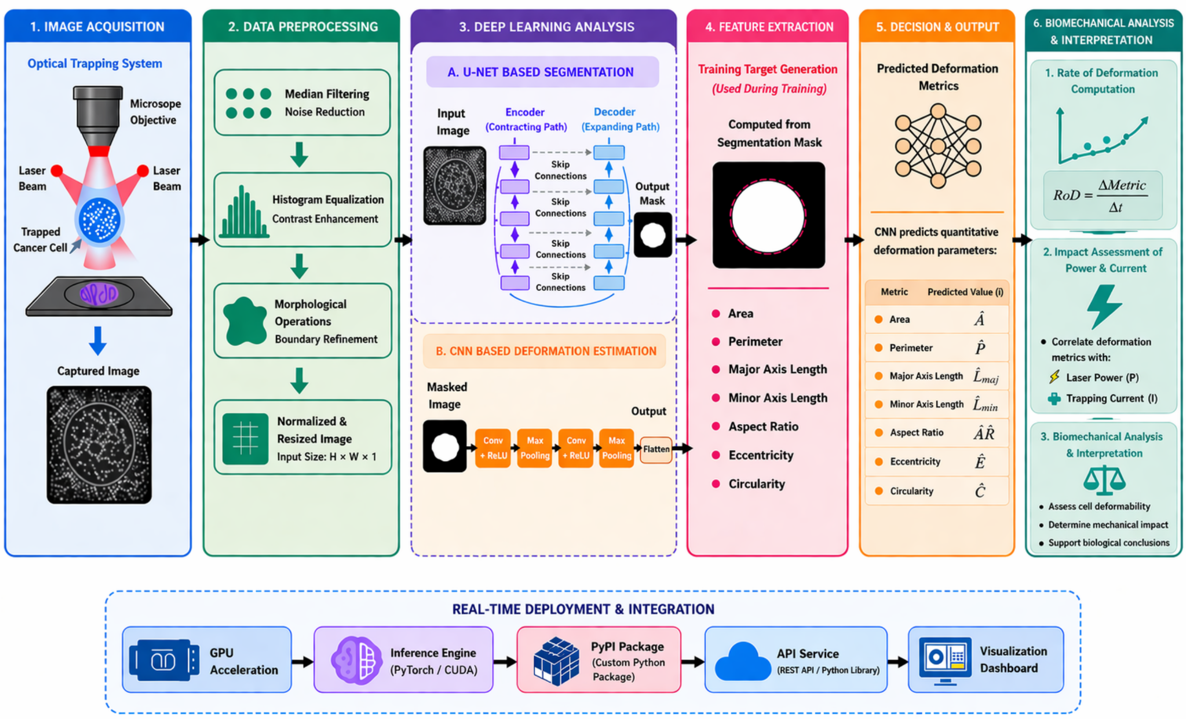

Figure 6: Proposed deep learning framework architecture.

At the input stage, raw cell images are acquired from the optical trapping imaging setup under different deformation conditions. These images serve as the primary input to the framework. Since raw microscopy images may contain noise, uneven illumination, and low contrast, they are first passed through a preprocessing block. The preprocessing stage is responsible for enhancing image quality and standardizing the data before analysis. Techniques such as median filtering, histogram equalization, and morphological operations are applied to suppress noise, improve contrast, and refine cell structures.

Following preprocessing, the system branches into two analytical pathways. The first pathway is the baseline CNN-based deformation estimation module, which performs direct learning from the processed images. This module has already been implemented using GPU acceleration and serves as the initial proof of concept for the feasibility of deep learning in this application. The CNN extracts hierarchical image features automatically and maps them to deformation-related outputs. Although this direct estimation approach provides a useful benchmark, it may not sufficiently capture precise boundary information needed for accurate quantitative interpretation.

To improve spatial localization and cell-region extraction, the second pathway introduces a U-Net-based image segmentation module. The purpose of this block is to isolate the cell from the image background and generate a segmentation mask that accurately represents the cell boundary. The segmented cell region is then forwarded to the deformation analysis stage, where quantitative geometric and morphological properties can be computed. These properties form the basis for more interpretable deformation characterization.

The deformation analysis block uses either segmented masks or ground truth contours to estimate deformation-related parameters. Typical descriptors include cell area, perimeter, aspect ratio, and eccentricity. These parameters provide a quantitative representation of the cell shape under optical force and serve as meaningful indicators of biomechanical response. By transforming image data into measurable deformation features, this stage links the deep learning pipeline with the physical interpretation of cell deformation.

The final block of the architecture is the decision or estimation module. In the baseline pathway, the CNN directly produces deformation-related predictions. In the segmentation-enhanced pathway, the extracted quantitative features are used to estimate deformation more explicitly. This design supports comparative analysis between direct image-based prediction and feature-assisted estimation, making it possible to assess the strengths and limitations of each approach.

The overall system architecture is intentionally designed to support future real-time integration. Since the framework follows a staged and modular structure, each component can be optimized for deployment in an automated optical trapping environment. This makes the proposed system suitable not only for offline experimental analysis but also for future development into an intelligent tool for real-time cell deformation monitoring and characterization.

3.3 Experimental Setup and Data Acquisition

The dataset used in this study is obtained from an optical trapping imaging system designed to capture deformation behavior of cancer cells under controlled optical forces. The setup consists of a dual-beam optical tweezer system integrated with a high-resolution microscopy imaging unit. Laser beams are used to trap and manipulate individual cells, while a microscope objective captures time-resolved images of the deformation process as depicted in Figure (7).

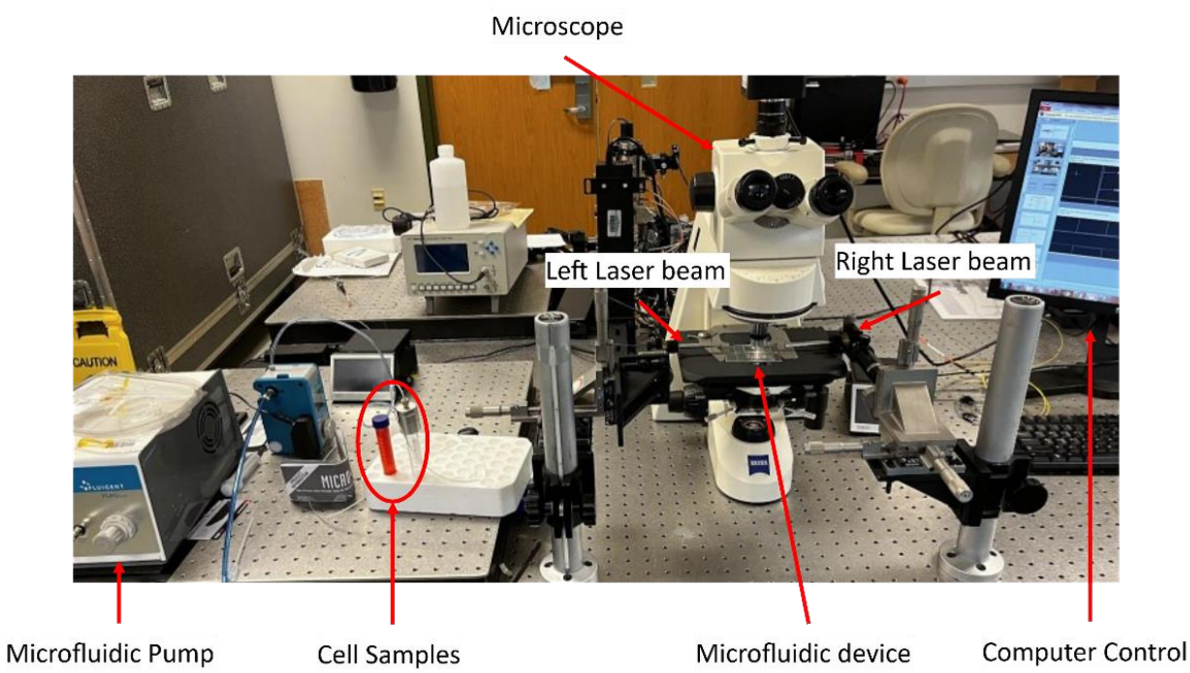

Figure 7: Dual-beam optical tweezer microfluidic setup [1].

During experimentation, cancer cells are subjected to varying levels of optical power and trapping current, which induce mechanical deformation. As the applied optical force increases, the cell undergoes changes in shape that are indicative of its biomechanical properties. These deformation sequences are recorded as grayscale image frames, forming the primary dataset for this work.

The acquired images exhibit variability in several aspects, including cell position, deformation magnitude, and background conditions. Unlike prior studies that assume centrally located cells, this work considers images where cells may appear at different spatial locations within the field of view. This introduces additional complexity but improves the robustness and generalizability of the proposed framework.

To ensure sufficient resolution for accurate deformation analysis, images are captured at higher pixel densities compared to existing approaches. This enables more precise segmentation and feature extraction, particularly for boundary-sensitive metrics such as perimeter and eccentricity.

The dataset consists of sequences of images captured over time, allowing for temporal analysis of deformation behavior. Each sequence corresponds to a specific experimental condition defined by laser power and trapping current settings. These parameters are recorded and later used in the biomechanical analysis stage to evaluate the relationship between applied optical forces and observed deformation.

Overall, the data acquisition process is designed to capture realistic experimental conditions while introducing variability that supports the development of a robust and deployable deep learning framework for quantitative cancer cell deformation analysis.

3.4 Data Preprocessing

The data processing stage converts raw optical trapping videos into structured image datasets suitable for deep learning and quantitative analysis. Since cancer cell deformation is captured as a time-varying sequence, the first step involves extracting representative frames from each recorded experiment in a controlled and reproducible manner.

In the current implementation, frame extraction is performed using a Python-based script developed with OpenCV and executed on the Anvil High Performance Computing

(HPC) platform by Purdue University. The use of HPC resources enables efficient handling of large video datasets and supports scalable preprocessing across multiple experimental samples.

The implemented framework supports two modes of temporal sampling. In the first mode, frames are extracted at fixed intervals by selecting every n-th frame, which reduces redundancy in high-frame-rate recordings. In the second mode, frames are sampled at a specified target frame rate, allowing alignment with a uniform temporal grid. This flexibility ensures that the extraction process can be adapted based on the deformation dynamics and storage requirements of different experiments.

Each selected frame is resized to a predefined spatial resolution prior to storage. This resizing step ensures consistency in input dimensions for subsequent CNN training. Furthermore, the proposed study aims to improve upon prior work by incorporating higher-resolution images, thereby preserving more spatial detail of the cell deformation process.

The extracted frames are stored in standard image formats such as PNG or TIFF. In addition, metadata associated with each frame is recorded, including the source video name, frame index, timestamp, file path, and output dimensions. This information is organized into a structured CSV file, providing traceability between the processed images and their corresponding temporal positions in the original video sequence. Such structured data management facilitates reproducibility and enables synchronization with downstream analysis tasks.

At the proposal stage, this frame extraction and formatting procedure represents the foundational component of the data processing pipeline. Building on this baseline, the proposed framework will incorporate additional preprocessing stages, including image enhancement, intensity normalization, and noise reduction. Furthermore, the framework will introduce spatially robust processing to handle cells located at arbitrary positions within the image frame, addressing a key limitation of existing approaches that assume centered cell positioning.

Future extensions will also include automated cell localization, segmentation, and feature refinement to improve the quality of inputs provided to the CNN model. These enhancements are expected to increase the robustness, generalization capability, and real-time applicability of the proposed deep learning framework for quantitative cancer cell deformation analysis.

In summary, the current implementation focuses on efficient frame extraction, resizing, and metadata generation from optical trapping videos. The proposed research extends this baseline by integrating advanced preprocessing, spatial variability handling, and segmentation-aware preparation to enable accurate and real-time deformation analysis in practical optical tweezer systems.

3.5 Ground Truth Deformation Analysis

To generate reference segmentation masks and quantitative deformation labels for supervised learning, an initial image-processing pipeline was developed in MATLAB Anvil HPC platform. This stage is not intended to serve as the primary segmentation method in the proposed framework; rather, it is used to construct a limited set of reliable ground-truth annotations for training the U-Net model.

The pipeline begins by loading processed cell images and converting them to grayscale where necessary. Each image is smoothed using a Gaussian filter to suppress noise and minor intensity fluctuations. Contrast enhancement is then performed using adaptive histogram equalization to improve the visibility of cell boundaries.

Subsequently, adaptive thresholding is applied to obtain a binary image representation. Let $I(x, y)$ denote the preprocessed grayscale image. A locally adaptive threshold $T(x, y)$ is computed such that the binary mask $B(x, y)$ is defined as

$$B(x, y) = \begin{cases} 1, & \text{if } I(x, y) > T(x, y), \\ 0, & \text{otherwise.} \end{cases} \tag{7}$$

To refine the segmentation, a sequence of morphological operations is applied, including removal of small objects, hole filling, and morphological opening and closing. These operations help eliminate noise and enforce spatial consistency in the segmented regions.

Connected-component analysis is then performed on the binary image. Since the experimental setup typically produces images where the target cell is located near the center of the field of view, the segmented object whose centroid is closest to the image center is selected as the candidate cell region. Let $\mathbf{c}_i = (x_i, y_i)$ denote the centroid of the i-th connected component and $\mathbf{c}_0$ the image center. The selected component is given by

$$i^* = \arg\min_i \|\mathbf{c}_i - \mathbf{c}_0\|_2 . \tag{8}$$

Additionally, area-based filtering is applied to discard components that are unrealistically small or excessively large.

From the final binary mask, a set of geometric deformation features is extracted using region-based analysis. These include:

- **Area** (A): number of pixels within the segmented region,
- **Perimeter** (P): length of the boundary of the region,
- **Centroid** (C_x, C_y): geometric center of the region,
- **Eccentricity** (e): measure of elongation of the fitted ellipse,
- **Major axis length** (L_{maj}) and **minor axis length** (L_{min}),
- **Aspect ratio** (AR):
$$AR = \frac{L_{\text{maj}}}{L_{\text{min}}}, \tag{9}$$
- **Solidity** (S): ratio of region area to its convex hull area,
- **Orientation** (θ): angle of the major axis with respect to the horizontal axis.

The generated binary masks are saved as image files, while the extracted features are exported into a structured dataset for further analysis and model development.

It should be noted that this preliminary segmentation approach assumes that the target cell is approximately centered in the image. While this assumption is suitable for constructing a curated ground-truth dataset, it is not valid for more general scenarios where cells may appear at arbitrary spatial locations.

In the proposed framework, this conventional segmentation pipeline will be used solely for generating ground-truth annotations. The primary segmentation task will be performed by a U-Net model, which is expected to provide improved robustness and generalization across variations in cell position, scale, and deformation patterns.

3.6 U-Net for Image Segmentation

Although the current framework utilizes a semi-automated, center-based segmentation approach implemented in MATLAB to generate ground truth masks, this method is limited in its scalability and robustness. It assumes that cells are centrally located and requires manual intervention, making it unsuitable for large-scale datasets and real-time deployment. Furthermore, variations in cell position, illumination, and imaging conditions can significantly degrade segmentation accuracy.

To address these limitations, this work proposes the adoption of a U-Net-based CNN for automated cell segmentation. U-Net is a widely used architecture in biomedical image analysis due to its ability to perform precise pixel-level segmentation while requiring relatively small training datasets. Its encoder–decoder structure, combined with skip connections, enables the network to capture both contextual and spatial information, making it particularly suitable for segmenting deformable biological structures such as cancer cells.

3.6.1 Training of the U-Net Model. In the proposed framework, the semi-automated masks generated using MATLAB will serve as ground truth labels for training the U-Net model. The corresponding optical trapping images will be used as input data. Once trained, the U-Net model will be capable of producing accurate segmentation masks for unseen images without manual intervention. These predicted masks will subsequently be used for the extraction of quantitative deformation features, including area, perimeter, eccentricity, and aspect ratio.

3.6.2 Significance of the U-Net Segmentation. The adoption of U-Net is expected to significantly improve segmentation accuracy and robustness across varying imaging conditions. It eliminates the need for manual intervention, enhances generalization to off-center and irregularly shaped cells, and enables scalable processing of large datasets. This is a critical step toward achieving real-time deployment in optical tweezer systems.

3.6.3 Evaluation Metrics for the U-Net Model. To quantitatively evaluate the performance of the proposed U-Net segmentation model, standard pixel-wise evaluation metrics commonly used in biomedical image analysis will be employed. These metrics assess the similarity between the predicted segmentation mask and the ground truth mask generated from the semi-automated MATLAB pipeline.

Let TP, FP, TN, and FN denote the number of true positive, false positive, true negative, and false negative pixels, respectively. The Dice Similarity Coefficient measures the overlap between the predicted mask and the ground truth mask. The Dice coefficient ranges from 0 to 1, where 1 indicates perfect overlap. It is defined as:

$$\mathrm{DSC} = \frac{2TP}{2TP + FP + FN} \tag{10}$$

Also known as the Jaccard index, the IoU evaluates the ratio of the intersection to the union of the predicted and ground truth masks. Higher IoU values indicate better segmentation performance.

$$\mathrm{IoU} = \frac{TP}{TP + FP + FN} \tag{11}$$

Precision measures the proportion of correctly predicted positive pixels out of all predicted positives:

$$\mathrm{Precision} = \frac{TP}{TP + FP} \tag{12}$$

Recall measures the proportion of actual positive pixels that are correctly identified:

$$\mathrm{Recall} = \frac{TP}{TP + FN} \tag{13}$$

Accuracy evaluates the overall correctness of the segmentation:

$$\mathrm{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN} \tag{14}$$

3.7 CNN-Based Regression Model for Cell Deformation Analysis

A CNN was developed to predict quantitative deformation parameters directly from optical trapping images. Unlike classification-based CNNs, this model is formulated as a regression problem, where the network predicts continuous biomechanical features extracted from segmented cell images.

3.7.1 Dataset and Input Representation. The dataset consists of cell images and corresponding deformation features obtained from a semi-automated segmentation pipeline. Each sample is represented as:

$$I \in \mathbf{R}^{H\times W\times 3} \tag{15}$$

where H and W denote the spatial dimensions and the three channels correspond to RGB images. The target variable for this study is the *Aspect Ratio*, defined in Equation (6). The dataset is split into training and validation sets using an 80/20 ratio.

3.7.2 Data Preprocessing. To improve generalization, the following preprocessing steps are applied:

- Image resizing to 224×224;
- Random horizontal flipping;
- Random rotation (up to $\pm 10°$); and
- Normalization using ImageNet statistics.

These transformations enhance robustness to variations in cell orientation and positioning.

3.7.3 Network Architecture. The proposed CNN architecture, referred to as *MegaNet*, consists of three convolutional blocks followed by a regression head as shown in Figure (8).

*3.7.3.1 Feature Extraction Layers.*Each convolutional block consists of convolution, ReLU activation, and max pooling:

$$\mathbf{F}_k = \mathrm{ReLU}(\mathbf{W}_k * \mathbf{X} + b_k) \tag{16}$$

The architecture is defined as:

- Conv($3 \to 16$) + ReLU + MaxPool
- Conv($16 \to 32$) + ReLU + MaxPool
- Conv($32 \to 64$) + ReLU + MaxPool

To ensure compatibility with varying input sizes, a Global Average Pooling (GAP) layer was applied:

$$\mathbf{z}_c = \frac{1}{H \times W} \sum_{i=1}^{H} \sum_{j=1}^{W} x_{c,i,j} \tag{17}$$

This produces a fixed-length feature vector of size 64.

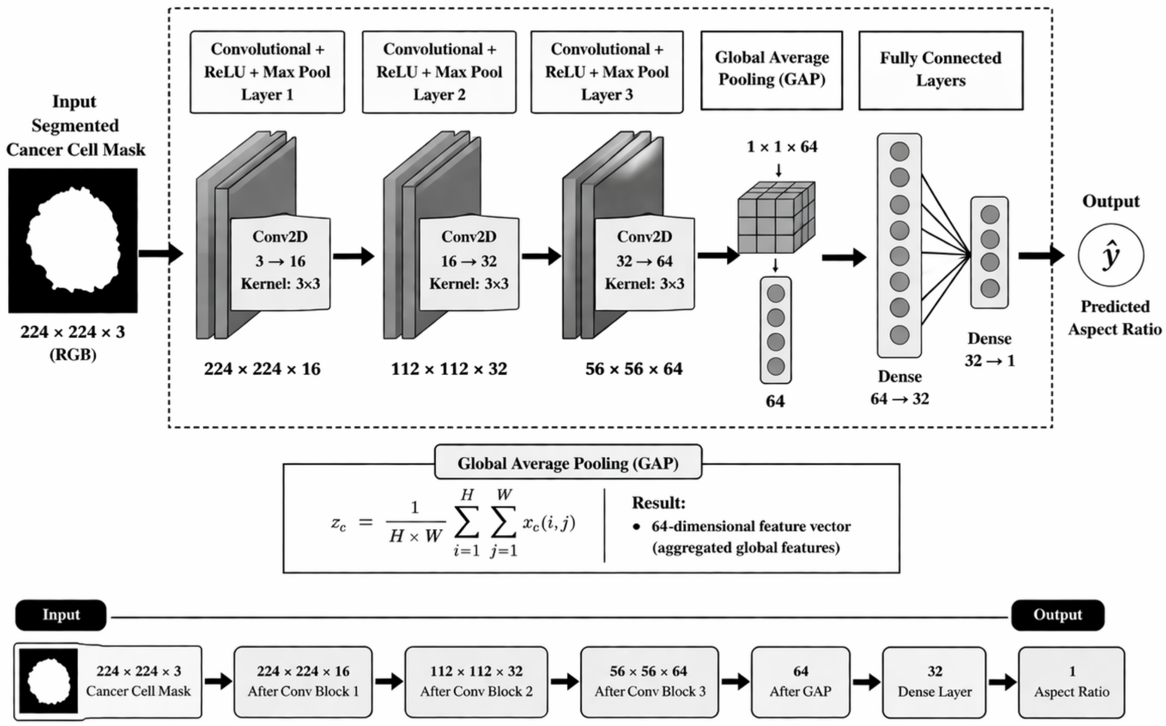

Figure 8: Proposed CNN architecture. The network takes segmented cell masks as input and extracts hierarchical features through three convolutional blocks (Conv + ReLU + Max Pool). GAP is used to obtain a compact feature representation, which is then passed through fully connected layers to predict the aspect ratio.

*3.7.3.2 Regression Head.*The extracted features are passed through fully connected layers:

$$\hat{y} = f(\mathbf{x}) = \mathbf{W}_2 \cdot \sigma(\mathbf{W}_1 \mathbf{x} + \mathbf{b}_1) + \mathbf{b}_2 \tag{18}$$

where the output is a single scalar representing the predicted aspect ratio.

3.7.4 Training Strategies and Optimizations. The model was trained using Mean Squared Error (MSE) loss:

$$\mathcal{L}_{\text{MSE}} = \frac{1}{N}\sum_{i=1}^{N}(y_i - \hat{y}_i)^2 \tag{19}$$

The AdamW optimizer was used for optimizing the training weights with weight decay regularization to prevent overfitting.

$$\theta_{t+1} = \theta_t - \alpha\frac{\hat{m}_t}{\sqrt{\hat{v}_t} + \epsilon} \tag{20}$$

3.7.5 Evaluation Metrics. Model performance was evaluated using:

- Mean Absolute Error (MAE)

$$MAE = \frac{1}{N}\sum_{i=1}^{N}|y_i - \hat{y}_i| \tag{21}$$

- Coefficient of Determination (R^2)

$$R^2 = 1 - \frac{\sum_{i=1}^{N}(y_i - \hat{y}_i)^2}{\sum_{i=1}^{N}(y_i - \bar{y})^2} \tag{22}$$

In summary, it is important to note that this CNN architecture represents a preliminary model intended to establish a baseline for automated deformation analysis of cancer cells. While the model demonstrates the feasibility of learning deformation-related features directly from segmented cell masks, its performance is constrained by factors such as limited dataset size and simplified architectural design. Future work will focus on improving the robustness and predictive capability of the model through architectural enhancements, expansion of the dataset, and incorporation of multi-parameter regression.

4. PRELIMINARY RESULTS

4.1 Overview

This section presents the initial results obtained from the proposed CNN-based regression model for cancer cell deformation analysis. The objective of this preliminary evaluation is to assess the feasibility of predicting quantitative deformation parameters directly from segmented cell images.

The model was trained on a dataset consisting of 83 labeled cell images, with an 80/20 split for training and validation. The target variable for this experiment was the aspect ratio, derived from the segmented cell geometry. Training was performed over 100 epochs using the AdamW optimizer and mean squared error (MSE) loss.

Results indicate that the model is capable of learning meaningful relationships between image features and deformation characteristics. A steady decrease in both training and validation loss was observed throughout the training process, accompanied by a corresponding reduction in mean absolute error (MAE).

The best validation performance achieved was:

- Mean Absolute Error (MAE): 0.512
- Mean Squared Error (MSE): 0.410
- Coefficient of Determination (R^2): -0.021

While the decreasing error metrics suggest that the model is learning relevant feature representations, the negative R^2 value indicates that the predictive performance is still limited and does not yet generalize well to unseen data. This is primarily attributed to the small dataset size and the complexity of the deformation patterns.

A regression analysis of predicted versus true aspect ratio values shows partial alignment with the ideal linear relationship, with noticeable deviations for certain samples. These results highlight both the potential and the current limitations of the proposed approach.

Overall, the preliminary results demonstrate the feasibility of using CNN for deformation prediction, while also emphasizing the need for further model refinement, larger datasets, and improved feature extraction through a U-Net model.

4.2 Experimental Observations

4.3 Image Preprocessing and Segmentation Results

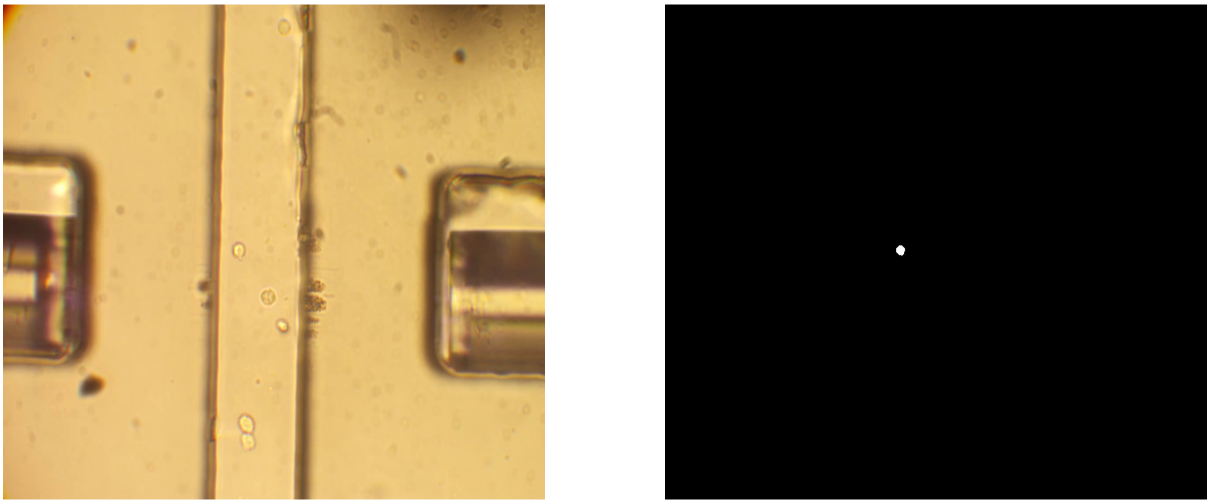

Figure 9: Image preprocessing and segmentation. Left: raw optical trapping image of cancer cells within a microfluidic channel. Right: corresponding binary mask obtained using MATLAB-based segmentation, highlighting the detected cell region used for feature extraction.

Figure (9) illustrates an example of the preprocessing and segmentation pipeline. The raw optical trapping image is first processed to enhance contrast and isolate regions of interest. A semi-automated MATLAB-based segmentation approach is then applied to generate binary masks representing the cell boundaries.

The resulting mask highlights the detected cell region used for extracting geometric features such as area, perimeter, and aspect ratio as shown in Table (1). While the segmentation method successfully identifies candidate regions, it exhibits limitations in accurately capturing full cell boundaries, particularly in cases with low contrast or overlapping structures.

Table 1: Sample of extracted morphological features from segmented cancer cell masks.

Filename	Area	Perimeter	C_x	C_y	Eccentricity	Major Axis	Minor Axis	Aspect Ratio	Solidity	Orientation
input_s...	88	31.088	267.5341	279.1477	0.4910	11.4290	9.9564	1.1479	0.9778	-81.2450
input_s...	90	31.642	266.7889	292.6333	0.6449	12.3402	9.4310	1.3085	0.9890	-88.7076
input_s...	95	33.023	265.6842	304.5263	0.6895	13.0854	9.4780	1.3806	1.0000	84.1248
input_s...	97	33.511	264.8866	312.5464	0.6442	12.9357	9.8943	1.3074	0.9898	80.1444

These limitations introduce potential inaccuracies in the ground truth labels used for training the CNN model, thereby affecting overall prediction performance. This observation motivates the need for a more robust, learning-based segmentation approach, such as U-Net, to improve boundary detection and feature consistency.

4.4 Performance of the CNN Regression Model

This section presents the preliminary evaluation of the proposed CNN for predicting the aspect ratio of cancer cells from segmented masks. The performance is assessed using three key visualizations: training and validation loss curves, predicted versus true regression analysis, and comparison with an ideal regression model.

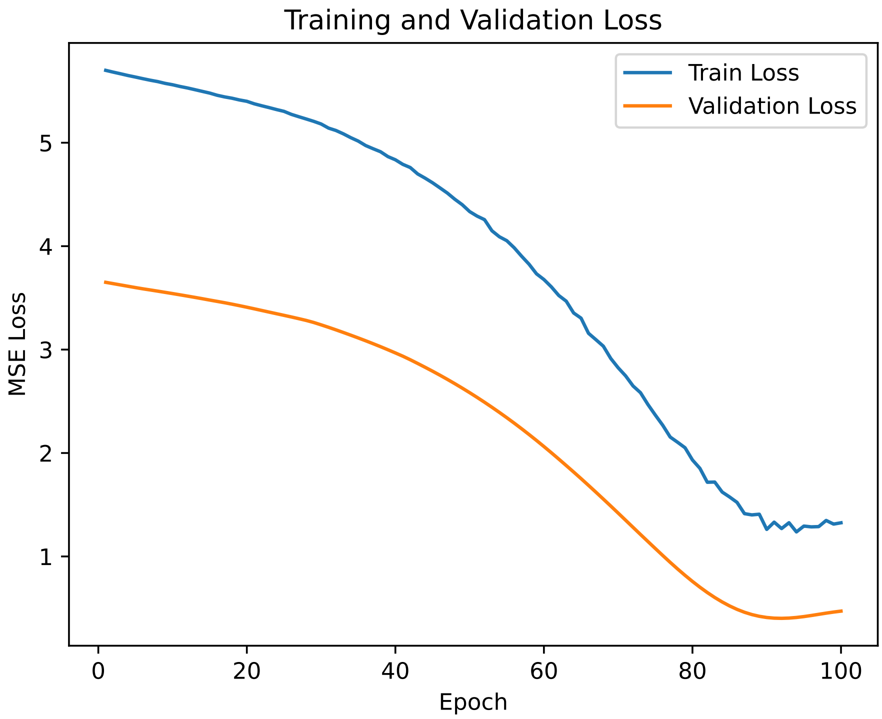

Figure 10: Training and validation loss curves of the CNN regression model over 100 epochs.

4.4.1 Training and Validation Loss. Fig. 10 shows the evolution of the training and validation loss during the learning process. Both curves exhibit a consistent decrease in mean squared error (MSE), indicating that the network is able to learn meaningful representations from the input data. The relatively close behavior of the training and validation curves suggests that severe overfitting is not present at this stage.

However, despite the reduction in loss, the magnitude of the error remains relatively high, indicating that the learned representation is still insufficient for accurate prediction of the deformation parameter.

4.4.2 Predicted Versus True Regression Analysis. The regression performance of the model is further illustrated in Fig. 11, which compares the predicted aspect ratio values with the corresponding ground-truth values. Ideally, predictions should align along a linear trend, reflecting accurate mapping between input features and the target variable.

In the current results, the predicted values are concentrated within a narrow range, while the true values exhibit a broader distribution. This indicates that the model is not fully capturing the variability in the deformation parameter and is biased toward average predictions. Such behavior is characteristic of underfitting and limited feature expressiveness.

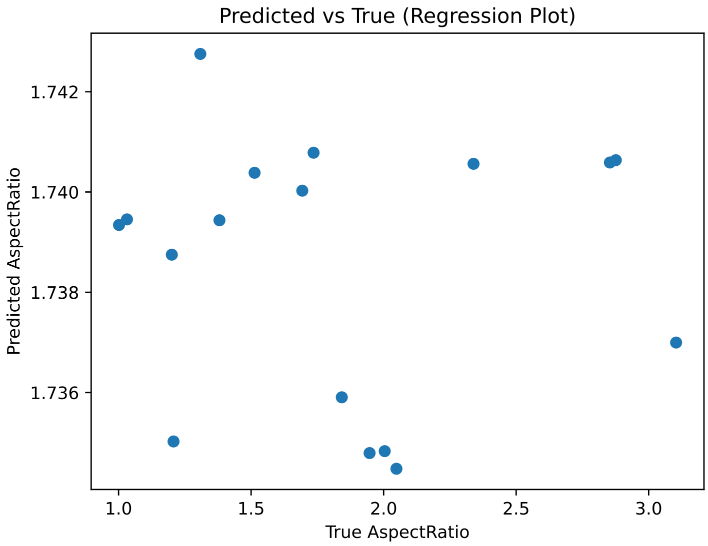

Figure 11: Predicted versus true aspect ratio values obtained from the CNN regression model.

4.4.3 Comparison with Ideal Regression. To provide a reference for optimal performance, Fig. 12 illustrates the ideal regression scenario in which all predicted values lie on the line $y = x$. Comparing Fig. 11 with this ideal case highlights the gap between the current model predictions and the desired one-to-one correspondence.

4.4.4 Discussion of Limitations. The observed limitations in prediction accuracy can be attributed to several factors. First, the dataset used for training is relatively small, which restricts the model's ability to generalize across diverse deformation patterns. Second, the quality of the segmentation masks used as input plays a critical role in determining the accuracy of the extracted features.

As observed in the preprocessing stage, the MATLAB-based segmentation approach does not always capture complete cell boundaries, leading to incomplete or noisy representations of the cell geometry. Consequently, the derived ground-truth labels (e.g., aspect ratio) may contain inaccuracies, which directly affect the learning process of

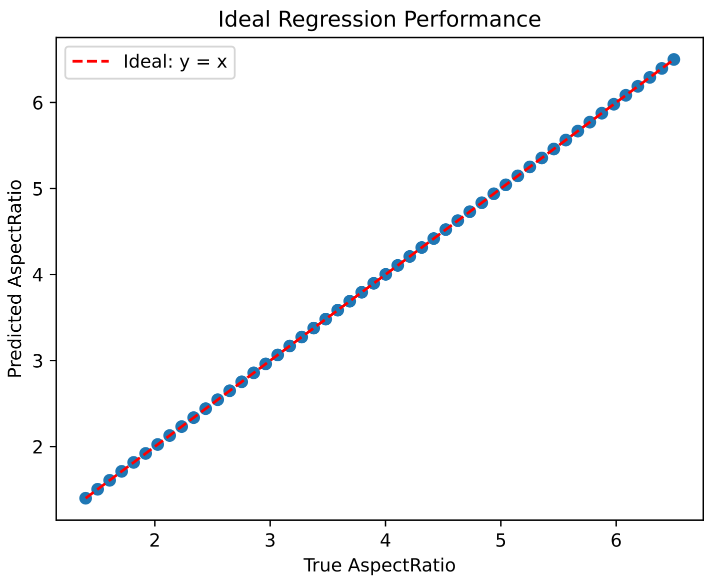

Figure 12: Ideal regression performance where predictions perfectly match ground-truth values ($y = x$).

the CNN. The current CNN architecture operates directly on segmented masks without explicitly learning spatial boundary features, limiting its ability to extract fine-grained deformation characteristics.

Overall, the preliminary results demonstrate that the proposed CNN model is capable of learning basic deformation patterns from segmented images. However, the current performance is limited by dataset size, segmentation quality, and model capacity. The integration of advanced segmentation techniques such as U-Net, along with architectural refinements and expanded datasets, is expected to significantly improve prediction accuracy in the main study.

5. CONCLUSIONS AND FUTURE WORKS

To address the limitations observed in the preliminary results, several improvements are proposed in the main study, spanning data acquisition, preprocessing, segmentation, and model design.

First, a larger and more diverse dataset will be acquired through controlled optical trapping experiments. Increasing the number of samples and capturing a wider range of deformation patterns will significantly improve the generalization capability of the model and reduce bias toward mean predictions.

Second, a more robust preprocessing pipeline will be implemented to enhance image quality and consistency prior to feature extraction. This includes median filtering for noise reduction, histogram equalization for contrast enhancement, morphological operations for refining structural features, and normalization and resizing to ensure uniform input representation. These steps are expected to improve the visibility and consistency of cell boundaries, thereby enabling more reliable feature learning.

Third, the current MATLAB-based segmentation approach will be semi-replaced with a deep learning-based method using U-Net. A small subset of the former will serve as the ground truth for training the U-Net model after proper manual inspection. Unlike conventional thresholding techniques, U-Net performs pixel-level semantic segmentation, allowing for accurate and consistent delineation of complex cell boundaries. This will significantly improve the quality of the extracted geometric features, such as area, perimeter, and aspect ratio, and reduce errors introduced by incomplete or noisy masks.

In addition to segmentation improvements, the CNN model will be extended from a single-output regression framework to a multi-output regression model. Instead of predicting only the aspect ratio, the enhanced model will simultaneously estimate multiple deformation-related parameters, including area, perimeter, eccentricity, and other morphological descriptors. This multi-task learning approach is expected to improve feature representation and enable a more comprehensive characterization of cell deformation.

Overall, these improvements aim to establish a more robust and scalable pipeline for deformation analysis. By combining enhanced data quality, advanced preprocessing, accurate segmentation, and a more expressive regression model, the proposed framework is expected to achieve significantly improved predictive performance in the main study.

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APPENDIX A

Thesis Timeline

Figure 13: Proposed thesis timeline from February to December 2026.